Functional Annotation Report: *soxA* (PP_0325 / Q88R09) — α-Subunit of Heterotetrameric Sarcosine Oxidase in *Pseudomonas putida* KT2440

Summary

The gene **soxA** (ordered locus **PP_0325**; UniProt **Q88R09**) of *Pseudomonas putida* strain KT2440 encodes the **~1004-residue α-subunit — the largest, catalytic subunit — of heterotetrameric sarcosine oxidase (TSOX; EC 1.5.3.1)**. TSOX is a soluble, cytoplasmic, bifunctional **diflavin metalloenzyme** built from four different polypeptides (α, β, γ, δ) that assemble into an αβγδ complex. The intact enzyme catalyzes the **oxidative demethylation of sarcosine (N-methylglycine) to glycine + hydrogen peroxide**, and, when tetrahydrofolate (THF) is present, **couples the removed one-carbon unit to the formation of 5,10-methylene-tetrahydrofolate**. This links the enzyme directly to folate-dependent one-carbon metabolism.

Critically, the specific role of the **α-subunit encoded by *soxA* is not the sarcosine-oxidation step itself** — that occurs at the noncovalent FAD site housed in the β-subunit. Instead, the α-subunit binds **NAD^+ and tetrahydrofolate** and carries out the **second half-reaction**: it is the **5,10-methylene-THF synthase module**, condensing the reactive channeled one-carbon iminium/formaldehyde intermediate onto folate and releasing glycine. This function resides in the C-terminal **GcvT / glycine-cleavage T-protein (aminomethyltransferase)-like folate-binding domain**, while an N-terminal Rossmann/glutathione-reductase-like **dinucleotide-binding domain** holds NAD^+. The two catalytic sites of the assembled enzyme lie ~35 Å apart and are connected by a large (~10,000 $Å^3$) internal cavity that channels the reactive intermediate from the β-subunit oxidation site to the α-subunit folate site — protecting the cell from a toxic, diffusible formaldehyde equivalent.

This assignment is supported by a convergent evidence chain: (1) the genomic context in KT2440, where PP_0325 sits within the canonical **glyA-soxBDAG-purU** folate/one-carbon operon; (2) crystallographic and biochemical studies of orthologous TSOX enzymes from *Pseudomonas maltophilia*, *Corynebacterium* sp. U-96 and P-1, and *Arthrobacter*, which localize

NAD^+ and folate to the α-subunit; and (3) quantitative full-length orthology (~47–48% identity across the entire ~1000-residue length) between Q88R09 and these structurally/biochemically characterized α-subunits, which justifies transferring their functional assignments to KT2440. The protein functions as a soluble cytoplasmic enzyme in **glycine-betaine / choline / creatine catabolism**, feeding one-carbon units into folate metabolism.

Key Findings

Finding 1 — *soxA* (PP_0325) is the α-subunit of heterotetrameric sarcosine oxidase, encoded in a conserved *glyA-soxBDAG-purU* operon

The genomic neighborhood of PP_0325 in *P. putida* KT2440 exactly recapitulates the canonical, well-characterized bacterial sarcosine-oxidase operon. Reading through the KEGG genome context: **PP_0322** = **glyA** (serine hydroxymethyltransferase) → **PP_0323** = **soxB (β)** → **PP_0324** = **soxD (δ)** → **PP_0325** = **soxA (α)** → **PP_0326** = **soxG (γ)** → **PP_0327** = **purU** (formyltetrahydrofolate deformylase) → **PP_0328** (formaldehyde dehydrogenase). PP_0325 spans 389,727–392,741 (3,015 bp ≈ 1,004 aa), matching the UniProt Q88R09 length of 1,004 residues exactly. This arrangement reproduces the classic **glyA-soxBDAG** gene order first described in *Corynebacterium* and *Arthrobacter*.

The original sequence analysis of the sarcosine oxidase operon established this canonical order: *"The sarcosine oxidase operon contains at least five closely packed genes encoding sarcosine oxidase subunits and serine hydroxymethyltransferase (glyA), arranged in the order glyAsoxBDAG"* (PMID: 7543100). A later study of glycine-betaine catabolic gene clusters confirmed the subunit assignment: *"the structural genes of heterotetrameric sarcosine oxidase (soxBDAG) and dimethylglycine dehydrogenase (dmg)"* (PMID: 11422368). Because the KT2440 locus order PP_0322–PP_0326 is identical to glyA-soxBDAG, PP_0325 is unambiguously **soxA (α)** — consistent with the UniProt gene name and its placement in the GcvT family.

Finding 2 — The assembled enzyme oxidatively demethylates sarcosine to glycine + H_2O_2 and couples the one-carbon unit to 5,10-methylene-THF (EC 1.5.3.1)

The overall chemistry of the enzyme complex of which *soxA* is a subunit has been established biochemically and structurally in orthologs from *Pseudomonas maltophilia* and *Corynebacterium*. TSOX contains **three coenzymes (FAD, FMN, NAD^+)** plus a tetrahydrofolate site and *"catalyzes the oxidation of sarcosine (N-methylglycine) to yield hydrogen peroxide, glycine and formaldehyde. In the presence of tetrahydrofolate, the oxidation of sarcosine is coupled to the formation of 5,10-methylenetetrahydrofolate"* (PMID: 16820168).

This bifunctionality — an amine-oxidation step joined to a folate one-carbon-transfer step — was described as coupling *"the oxidation of the methyl group in sarcosine (N-methylglycine) and transfer of the oxidized methyl group into the one-carbon metabolic pool"* (PMID: 15922624). The reactive iminium/formaldehyde intermediate produced by C–H bond cleavage at the FAD site is thus not simply released as free formaldehyde in the presence of folate; instead it is captured onto THF. This is the molecular basis of EC 1.5.3.1 and directly ties the enzyme to one-carbon (folate) metabolism.

Finding 3 — The α-subunit specifically houses NAD^+ and the tetrahydrofolate-binding site: it is the 5,10-methylene-THF synthase half of the enzyme

Multiple crystal structures resolve the enzyme's cofactors by subunit, and they consistently place the NAD^+ and folate sites in the **α-subunit**. In the 1.85 Å *P. maltophilia* structure, *"The NAD^+ and putative folate binding sites are located in the alpha-subunit. The FAD binding site is in the beta-subunit. FMN is bound at the interface of the alpha and beta-subunits"* (PMID: 16820168). The independent *Corynebacterium* sp. U-96 structure agrees: *"The alpha subunit is composed of two domains, contains NAD(+), and binds folinic acid"* (PMID: 15946648).

Functionally, the folate site in the α-subunit is where the second half-reaction occurs. Analysis of the channeling architecture in *Corynebacterium* U-96 states that *"The third channel goes through the α-subunit and has a folinic acid-binding site, where the iminium intermediate is converted to Gly and either formaldehyde or, 5,10-methylenetetrahydrofolate"* (PMID: 20675294). Biochemical subunit dissection provides orthogonal confirmation: *"The alpha subunit and the alphagamma complex were each found to contain 1 mol of NAD(+) but no FAD. Since NAD(+)*

binds to alpha, FAD probably binds to beta" (PMID: 11330998). Thus, the α-subunit encoded by *soxA* is the **NAD^+- and folate-binding, glycine-forming / methylene-THF-synthesizing module** of the enzyme — not the sarcosine dehydrogenase site.

Finding 4 — Domain architecture and evolutionary origin: an N-terminal Rossmann NAD^+-binding domain fused to a C-terminal GcvT/T-protein folate-binding domain

The α-subunit is a two-domain protein whose modular architecture reflects its two roles (dinucleotide binding and folate one-carbon transfer). Sequence analysis showed that the α-subunit *"contains a second ADP-binding motif within an approximately 280 residue region near the NH_2 terminus that exhibits homology with subunit A from octopine and nopaline oxidases"* (PMID: 7543100). Structurally, this N-terminal module resembles a flavoprotein FAD-binding fold repurposed for NAD^+: *"The N-terminal half of the alpha subunit of TSOX (alphaA) is closely similar to the FAD-binding domain of glutathione reductase but with NAD^+ replacing FAD"* (PMID: 16820168).

The ~380-residue C-terminal region is homologous to the **glycine-cleavage T-protein (GcvT / aminomethyltransferase)** and to the C-terminal half of dimethylglycine dehydrogenase — the folate-binding fold that carries out one-carbon transfer. This is fully consistent with the UniProt/InterPro annotation of Q88R09, which assigns it to the **GcvT family** and lists the FAD/NAD-binding domain (IPR023753, IPR036188) together with the GcvT/YgfZ/DmdA folate-binding domain (IPR028896, IPR029043) plus a 2Fe-2S-binding N-terminal signature (IPR042204). The bioinformatic domain composition therefore independently predicts exactly the two functions (NAD^+ binding + folate one-carbon chemistry) that structural biology localizes to this subunit.

Finding 5 — Pathway context and localization: a soluble cytoplasmic enzyme feeding one-carbon units into folate metabolism during glycine-betaine/choline catabolism

The *soxBDAG* operon is embedded within a **folate one-carbon gene cluster**. Immediately upstream lies **glyA** (serine hydroxymethyltransferase; serine + THF ⇌ glycine + 5,10-methylene-THF), and immediately downstream lie **purU** (10-formyl-THF deformylase) and a **formaldehyde dehydrogenase** (KT2440 PP_0322–PP_0328). This gene neighborhood positions sarcosine oxidase as a supplier of one-carbon units to the folate pool. The proposed physiological pathway is **glycine betaine → dimethylglycine → sarcosine → glycine**, with the liberated methyl-derived one-carbon groups assimilated via tetrahydrofolate: *"pathways are* *proposed for glycine betaine catabolism in Arthrobacter species, which involve the identified folate-dependent enzymes"* (PMID: 11422368). The original operon analysis emphasized that the cluster *"reveals homologies with key enzymes of folate one-carbon metabolism"* (PMID: 7543100).

Regarding localization, sarcosine oxidase is a classic **soluble cytoplasmic flavoenzyme**, and UniProt Q88R09 carries no annotated signal peptide or transmembrane segments, consistent with a cytoplasmic location where THF and NAD^+ are available.

Finding 6 — Substrate specificity and two-site organization: *soxA*/α is the folate half-site; sarcosine (N-methylglycine) is the primary substrate

The assembled enzyme has two catalytic sites ~35 Å apart: *"The sarcosine dehydrogenase and 5,10-methylenetetrahydrofolate synthase sites are 35 A apart but connected by a large internal cavity (approximately 10,000 A^3)"* (PMID: 16820168). Sarcosine oxidation occurs at the β-subunit FAD; the α-subunit (*soxA*) holds NAD^+ and the folate site (the methylene-THF synthase half-reaction). It is therefore essential to state clearly that *soxA* is **not** the sarcosine-oxidation site — but the substrate consumed by the overall enzyme is the secondary amino acid **sarcosine**.

Substrate-specificity data from the related monomeric sarcosine oxidase (MSOX) define the family's selectivity: MSOX oxidizes sarcosine best, with only minor activity on other secondary amino acids, and importantly *"MSOX can oxidize other secondary amino acids (N-methyl-L-alanine, N-ethylglycine, and L-proline), but N,N-dimethylglycine, a tertiary amine, is not a substrate"* (PMID: 10913293). This confirms that **sarcosine, not dimethylglycine, is the substrate of sarcosine oxidase** (dimethylglycine is instead handled by the separate dimethylglycine dehydrogenase, *dmg*, in the same catabolic pathway).

Finding 7 — KT2440 *soxA* (Q88R09) is a full-length ortholog (~47–48% identity) of structurally/biochemically characterized TSOX α-subunits, justifying functional transfer

Global (Needleman–Wunsch) alignment of Q88R09 (1,004 aa) against reviewed Swiss-Prot sarcosine-oxidase α-subunits gives high, full-length identity:

Ortholog (α-subunit)	Organism	UniProt	% Identity to Q88R09	Notes
Q46337	*Corynebacterium* sp. P-1	Q46337	**48.3%** (518/1072)	Source of biochemical NAD^+/α assignment
Q9AGP1	*Arthrobacter* sp.	Q9AGP1	47.9% (512/1068)	Glycine-betaine catabolism operon
Q50LF0	*Corynebacterium* sp. U-96	Q50LF0	47.5% (507/1067)	Crystal structure solved
O87386	*Sinorhizobium meliloti*	O87386	47.1%	Additional ortholog

The alignments span the full ~1,000-residue length, covering both the N-terminal NAD^+-binding domain and the C-terminal GcvT/folate domain. Because the U-96 (Q50LF0) and P-1 (Q46337) enzymes are exactly those for which crystal structures and subunit-resolved cofactor assignments were determined — *"The alpha subunit is composed of two domains, contains NAD(+), and binds folinic acid"* (PMID: 15946648) and *"The alpha subunit and the alphagamma complex were each found to contain 1 mol of NAD(+) but no FAD"* (PMID: 11330998) — the ~47–48% full-length identity provides a quantitative, defensible basis for transferring these functional assignments to the KT2440 protein.

Mechanistic Model / Interpretation

Heterotetrameric sarcosine oxidase is best understood as a **two-active-site molecular machine with substrate channeling**. The four subunits (α = *soxA*/PP_0325, β = *soxB*/PP_0323, γ = *soxG*/PP_0326, δ = *soxD*/PP_0324) assemble to co-locate three flavin/dinucleotide cofactors and a folate site.

```
                  SARCOSINE (N-methylglycine)
                           │
                           ▼
         ┌─────────────────────────────────────┐
         │  β-subunit (soxB)                     │
         │  noncovalent FAD  ── oxidation site   │  ← C–H bond cleavage,
         │  (2-furoate = competitive inhibitor)  │    O2 → H2O2
         └───────────────┬───────────────────────┘
                         │  reactive iminium /
                         │  formaldehyde equivalent
                         │  (5-oxazolidinone intermediate)
              ~35 Å internal cavity (~10,000 Å³)
              — substrate channeling, prevents
                escape of toxic formaldehyde —
                         │
                         ▼
         ┌─────────────────────────────────────┐
         │  α-subunit (soxA / PP_0325)          │
         │  N-term: Rossmann NAD+-binding domain │
         │  C-term: GcvT/T-protein folate domain │  ← 5,10-methylene-THF
         │  binds NAD+ + tetrahydrofolate        │    synthase half-reaction
         └───────────────┬───────────────────────┘
                         │
          ┌──────────────┴──────────────┐
          ▼                             ▼
    + THF: 5,10-methylene-THF      − THF: free formaldehyde
    + GLYCINE                      + GLYCINE
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The covalent FMN sits at the α/β interface and mediates electron transfer, functioning as an electron transferase between the two flavins; the δ (soxD) and γ (soxG) subunits complete the assembly and, in the case of the γ chain, associate with the α-subunit (the αγ complex retains NAD^+).

Physiologically, the enzyme operates in the middle of a linear methylamine-catabolic funnel in *P. putida*:

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glycine betaine ──────→ dimethylglycine ──(Dmg dehydrogenase)──→ sarcosine
                                                                     │
                                              SARCOSINE OXIDASE (soxBDAG)
                                                                     ▼
                                         GLYCINE  +  1-C unit on THF (5,10-CH2-THF)
                                                                     │
                         glyA (SHMT), purU (10-CHO-THF deformylase), FDH ┘
                                    (folate one-carbon pool)
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The **evolutionary logic** is elegant: the α-subunit is a fusion of two ancient one-carbon-metabolism modules — a dinucleotide-binding oxidoreductase fold (glutathione-reductase-like, here binding NAD^+ instead of FAD) and a GcvT/aminomethyltransferase folate-binding fold (the same fold used by the glycine-cleavage T-protein and dimethylglycine dehydrogenase). This fusion allowed sarcosine oxidase to internalize the folate-transfer step that, in the glycine cleavage system, is performed by a standalone T-protein. The net effect is that the reactive one-carbon intermediate never leaves the enzyme, protecting the cell from free formaldehyde while efficiently loading the folate pool.

Evidence Base

PMID	Title (abbrev.)	How it supports this annotation
7543100	*Sequence analysis of sarcosine oxidase and nearby genes...folate one-carbon metabolism*	Defines canonical **glyA-soxBDAG** operon order (→ identifies PP_0325 as soxA/α); describes N-terminal ADP-binding motif of α; links operon to folate one-carbon metabolism
16820168	*Heterotetrameric sarcosine oxidase: structure of a diflavin metalloenzyme at 1.85 Å*	Assigns **NAD^+ + folate to α**, FAD to β, FMN to α/β interface; overall reaction; two sites 35 Å apart, ~10,000 $Å^3$ cavity; α N-terminal = glutathione-reductase-like NAD^+ fold
15946648	*Crystal structure of heterotetrameric sarcosine oxidase from Corynebacterium U-96*	Independent structure: **α is two-domain, binds NAD^+ and folinic acid** (ortholog Q50LF0)
11330998	*Organization of coenzymes/ subunits and role of covalent flavin link*	Biochemical proof: α and αγ contain **1 mol NAD^+, no FAD** → NAD^+ binds α (ortholog Q46337)
20675294	*Channeling and conformational changes in TSOX from Corynebacterium U-96*	Localizes glycine-forming / methylene-THF-synthase step to the **α-subunit folinic-acid site**; iminium intermediate channeled
15922624	*Cloning, expression, crystallization of TSOX from P. maltophilia*	States bifunctional coupling of sarcosine methyl oxidation to the **one-carbon (folate) pool**
11422368	*Genes for dimethylglycine/ sarcosine degradation in Arthrobacter; glycine betaine catabolism*	Confirms **soxBDAG** = **four TSOX subunits**; places enzyme in glycine-betaine catabolism / folate assimilation
10913293	*Monomeric sarcosine oxidase kinetics with alternate substrates*	Family substrate specificity: **sarcosine primary substrate**, minor secondary amino acids, **N,N-dimethylglycine NOT a substrate**

Supporting/contextual literature reviewed includes mechanistic and structural studies of related enzymes: the flavin-thiolate (4a-sulfide) adduct of the TSOX FMN (PMID: 16934831); stopped-flow kinetics and quantum-tunneling C–H cleavage in *Arthrobacter* TSOX (PMID: 10684595); interflavin electron transfer in sarcosine oxidase (PMID: 1657156); umbrella-sampling analysis of intermediate (5-oxazolidinone) transport through the internal tunnels (PMID: 33576049); the structurally related FAD/FMN/ATP L-proline dehydrogenase complex whose β-subunit resembles monomeric sarcosine oxidase (PMID: 16027125); and MSOX substrate/mechanism studies (PMID: 16300392, PMID: 22432926). Engineering/specificity studies of sarcosine-oxidase family enzymes (PMID: 35124004, PMID: 40932061, PMID: 37968575, PMID: 24925096) reinforce the secondary-amino-acid/N-demethylase chemistry of the family.

Limitations and Knowledge Gaps

1. **No direct experimental characterization of the KT2440 protein.** All functional assignments for Q88R09 are transferred by orthology (~47–48% full-length identity) from TSOX enzymes of *P. maltophilia*, *Corynebacterium* sp. U-96 and P-1, and *Arthrobacter*. No published enzymology, crystal structure, knockout phenotype, or localization experiment specific to *P. putida* KT2440 PP_0325 was identified. The identity level, while high and full-length, is below the ~60–70% typically considered "safe" for automatic transfer of fine catalytic detail, so residue-level active-site claims should be treated as strong predictions rather than proven facts.
2. **Folate site is "putative" in the primary structure.** In the 1.85 Å *P. maltophilia* structure the folate site is described as the *"putative folate binding site"*; it is inferred from folinic-acid soaks in the *Corynebacterium* structures and from GcvT homology, not from a co-crystal with a physiological methylene-THF product.
3. **Physiological pathway is proposed, not proven in KT2440.** The glycine-betaine → dimethylglycine → sarcosine → glycine route and the operon-based one-carbon assimilation model derive from *Arthrobacter* and sequence/operon analysis. Whether *P. putida* KT2440 actually induces this operon under glycine-betaine/choline/creatine growth, and the exact upstream demethylase(s), have not been experimentally verified here.
4. **Subunit stoichiometry and assembly in KT2440 unconfirmed.** The αβγδ composition, covalent FMN attachment chemistry, and channeling behavior are established for the model orthologs but not directly demonstrated for the KT2440 complex.

5. **The 2Fe-2S signature (IPR042204)** appears in the InterPro annotation, but classical TSOX α-subunits are not described as iron-sulfur proteins in the biochemical literature reviewed; this domain match may reflect a distant fold relationship rather than a functional [2Fe-2S] cluster and warrants caution.

Proposed Follow-up Experiments / Actions

1. **Heterologous expression and cofactor analysis.** Clone the KT2440 *soxBDAG* operon (PP_0323–PP_0326) into *E. coli*, purify the complex, and confirm the predicted cofactor set (FAD, FMN, NAD^+) and the α-subunit's NAD^+/folate binding by spectroscopy and denaturation-release assays — directly replicating the PMID: 11330998 subunit dissection for the *P. putida* enzyme.
2. **Steady-state and stopped-flow kinetics.** Measure sarcosine oxidation (kcat, Km, H_2O_2 production) and THF-dependent 5,10-methylene-THF formation to quantitatively confirm EC 1.5.3.1 activity and folate coupling; test whether dimethylglycine is excluded as a substrate, as predicted from family specificity (PMID: 10913293).
3. **Gene deletion / growth phenotyping.** Construct a KT2440 ΔPP_0325 (and Δoperon) mutant and test growth on sarcosine, creatine, choline, and glycine betaine as sole carbon/nitrogen sources to establish the in vivo catabolic role and confirm operon function.
4. **Transcriptional induction analysis.** Use RT-qPCR or RNA-seq to determine whether the *glyA-soxBDAG-purU* cluster is co-induced during growth on glycine-betaine/choline/sarcosine, confirming it functions as a one-carbon catabolic operon in KT2440.
5. **Structure determination or AlphaFold-Multimer modeling.** Solve or model the KT2440 αβγδ complex to verify the two-site/channeling architecture and to map the α-subunit NAD^+ and folate sites at residue resolution, validating the ~35 Å / ~10,000 $Å^3$ cavity architecture predicted from orthologs.
6. **Resolve the 2Fe-2S annotation.** Test experimentally (UV-vis, EPR, iron content) whether the KT2440 complex contains any iron-sulfur center, to confirm or reject the IPR042204 domain match as functionally meaningful.

Conclusion

The gene *soxA* (PP_0325, Q88R09) of *P. putida* KT2440 encodes the **α-subunit of heterotetrameric sarcosine oxidase**, the enzyme's NAD^+- and tetrahydrofolate-binding catalytic module. It performs the **5,10-methylene-THF synthase half-reaction** — converting the channeled iminium/formaldehyde intermediate (generated at the β-subunit FAD sarcosine-oxidation site) into glycine plus a folate-bound one-carbon unit — via its C-terminal GcvT/T-protein-like domain, with an N-terminal Rossmann domain holding NAD^+. It is a **soluble cytoplasmic** enzyme functioning in **glycine-betaine/choline/creatine catabolism**, feeding one-carbon units into folate metabolism through the conserved *glyA-soxBDAG-purU* operon. All assignments rest on strong operon-context, structural, and biochemical evidence from ~47–48%-identical, full-length orthologs, but await direct experimental confirmation in *P. putida* KT2440 itself.